BIC21203 WEB DEVELOPMENT

Lab 4a JavaScript

Functions, events, validation, loops and DOM manipulation

Student name	Complete before submission
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Academic session	2026/2027
GitHub repository	Add after publishing

Report scope

This report demonstrates three working browser tasks: changing text through an event listener, validating numeric input with conditional logic, and generating list items with a loop and DOM methods.

Submission note: add the final public GitHub URL and personal details before uploading this report to AUTHOR.

Question 1 Interactive Text Changer

The page selects the paragraph and button by ID. The named **changeMessage()** function updates the paragraph through **textContent**, and **addEventListener()** connects the function to the click event.

Check	Observed result
Initial state	Welcome to my website!
After button click	Hello there!
Event method	addEventListener("click", changeMessage)

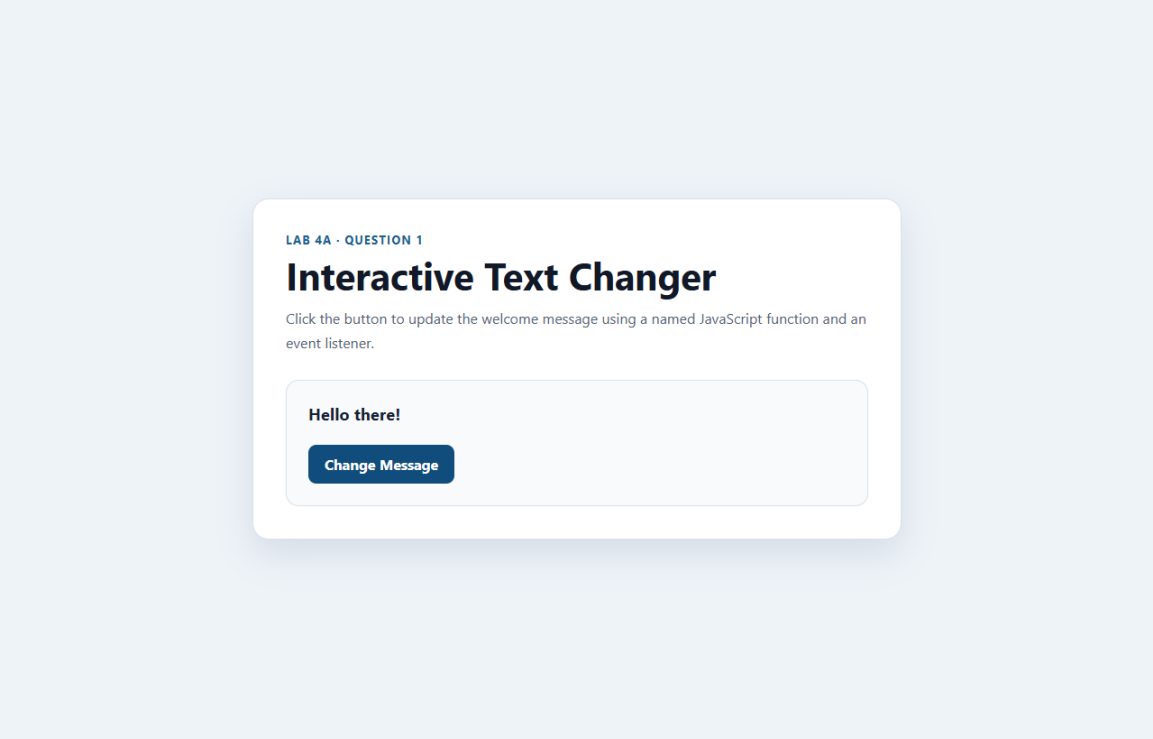


Figure 1. The required message after selecting Change Message.

Question 2 Smart Number Validator

The **checkInput()** function trims the input, rejects empty text, converts valid numeric strings with **Number()**, and rejects non-finite values. An if, else if, and else structure produces the required comparison message in the live status region.

Input	Expected category	Observed message	Result
5	Lower	The number is lower than 10	Pass
10	Equal	The number is equal to 10	Pass
15	Higher	The number is higher than 10	Pass
abc	Invalid	Please enter a valid number!	Pass
Empty	Invalid	Please enter a valid number!	Pass
10.5	Higher	The number is higher than 10	Pass

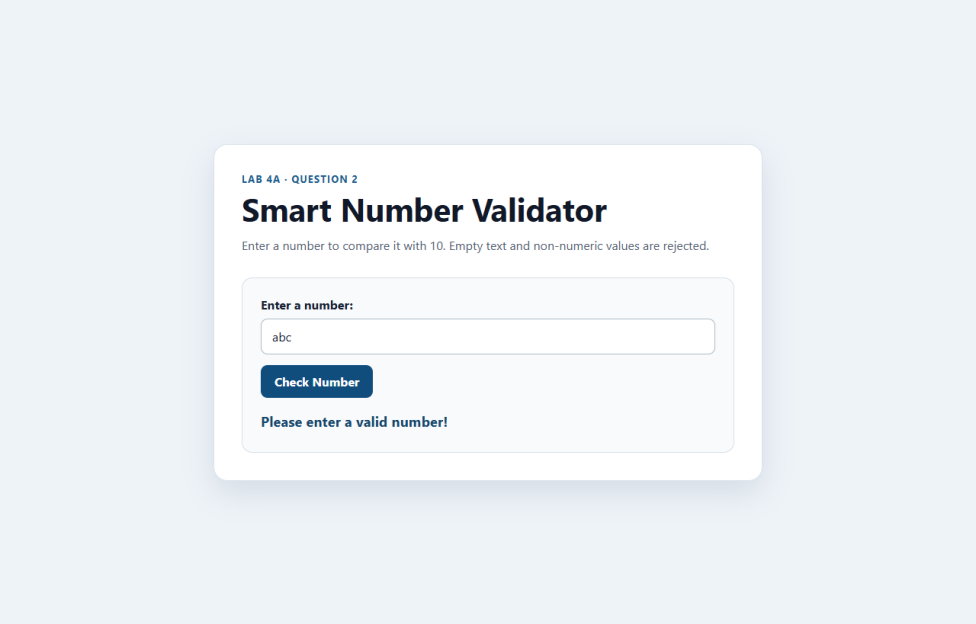


Figure 2. Invalid text is handled without inserting user input as HTML.

Question 3 Dynamic Number Generator

The **generateNumbers()** function first clears the ordered list with **replaceChildren()**. A for loop then creates five list items with **document.createElement()**, assigns each label through **textContent**, and appends each item to the list.

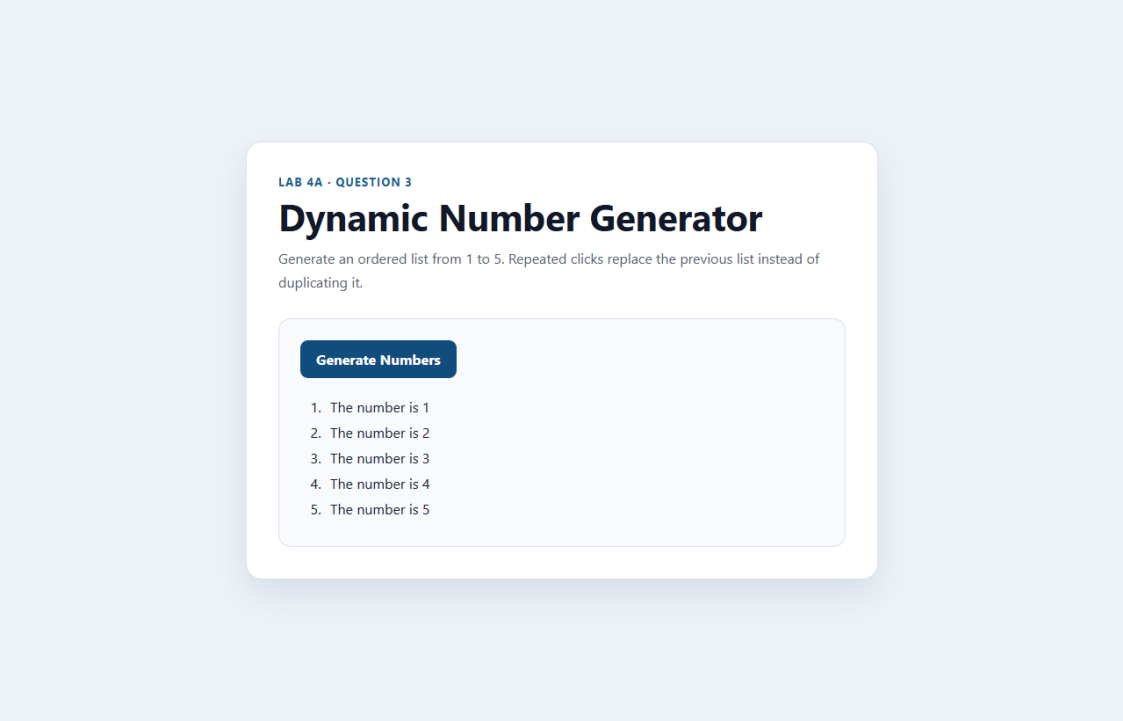


Figure 3. Exactly five generated items remain after repeated button clicks.

Verification

Automated browser checks selected Generate twice and counted five list items after both clicks. The first item was The number is 1 and the last item was The number is 5.

Reflection

This lab showed how small named functions make event-driven code easier to read and test. Input should be normalized before conversion so empty text is not treated as zero. DOM methods such as textContent and createElement provide safe, explicit page updates. Clearing existing output before a loop also makes repeated user actions predictable.